

Bence Jordan Danko

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EDUCATION

San Jose State University <i>M.Sc. in Applied Data Intelligence</i>	2024 – Present San Jose, USA
San Jose State University <i>B.Sc. in Computer Science</i>	2020 – 2024 San Jose, USA

PROJECTS

Edge Vision Smart Elevators: Privacy-Preserving Headcount Analysis

<https://github.com/bencejdanko/Edge-AI-People-Counting-NuMaker-X-M55M1D>

- Architected a privacy-preserving edge AI headcount system on the Nuvoton NuMicro M55M1 MCU, running private YOLOv8n-Relu6 people-detection inference on the Arm Ethos-U55 NPU and relaying counts to a remote server via Wi-Fi
- Achieved 25 FPS video at 1.5MB SRAM constraints via YOLOv8n-Relu6 and 38 FPS via FOMO on the Nuvoton NuMicro M551
- Engineered and deployed an INT8-quantized object detection pipeline optimized via the Vera compiler to run directly on the Arm Ethos-U55 NPU
- Published the open-source fomo-edge-ai/fomo PyPi package to streamline end-to-end training and processing of the FOMO vision model for Nuvoton using PyTorch and LiteRT for quantization.
- Developed low-level C firmware for the Cortex-M55, including UDP server and real-time inference handlers, and using FreeRTOS to optimize inference scheduling, screen updates and network count relays.
- Utilized Keil IDE and stack trace analysis to optimize memory allocation and resolve critical stack/heap bugs.
- Collaborated with Nuvoton engineers to validate system performance to demonstrate at their headquarters.

Multimodal API Agent for Warehouse Logistics

<https://github.com/bencejdanko/segmentation-ocr-multimodal-api-agent>

- Developed a multimodal API agent for warehouse logistics to automate inventory audits.
- Locally hosted APIs for SAM 3 and GLM-OCR using vLLM and FastAPI and the Transformers library on the NVIDIA Jetson AGX Orin, using vision segmentation, OCR and large language models for automated inventory management under 50W consumption.
- Implemented an orchestration pipeline that integrates SAM 3 segmentation with LLM-based filtering, allowing for dynamic re-prompting and precise localization of front-row product stacks.
- Built a comprehensive data extraction service that combines bounding box analysis and OCR processing to synthesize structured inventory reports, including product names, instance counts, and quantity validation.
- Engineered a dashboard for displaying real-time segmented crops, detected product counts, and extracted textual labels for warehouse management validation.

Mobilint MLA100 NPU Computer Vision Applications

<https://github.com/bencejdanko/image-restoration-convolution-mla100-optq-qat>

- Worked with Mobilint hardware and compiler engineers to evaluate computer vision applications for the MLA100 NPU, achieved <6ms processing latencies for 256x256 images.
- Replaced a high-latency ResNet baseline (400ms+) with a custom 0.2M parameter Convolutional Network by optimizing kernel launches and restricting feature maps to 64 to minimize hardware-induced latency.
- Orchestrated large-scale training and data processing jobs using modal.